

ANANTH KAMATH

+91 79756 46956 · ananthkamath916@gmail.com · linkedin.com/in/ananth-kamath77
github.com/Anantkamath00 · **Portfolio:** github.com/Anantkamath00/Ananth_Kamath

PROFESSIONAL SUMMARY

Detail-oriented cybersecurity researcher and software engineer specializing in safe identity framework design, defensive application development, and enterprise data protection architectures. Combines structured academic engineering methodologies with real-world threat validation models gained via corporate technology internships and competitive threat engineering tracks. Adept at establishing zero-trust pipeline mechanisms, minimizing web application exploit surfaces, and auditing cryptographic systems without introducing operational processing bottlenecks.

EDUCATION

Yenepoya (Deemed to be University) Mangaluru, India
Bachelor of Computer Applications (BCA) in Cyber Security, Ethical Hacking & Digital Forensics Jul 2023 – Jul 2026

- Developed in explicit academic collaboration with IBM, focusing heavily on modern information defense paradigms, secure operating systems, and defensive code analysis.

Sri Rama Pre University College Karnataka, India
Pre-University Course (PUC) in Science Jul 2020 – Jul 2022

TECHNICAL CORE COMPETENCIES

- **Identity & Access Management (IAM):** Zero-Trust System Engineering, Multi-Factor Authentication Architecture (MFA), Time-Based One-Time Password (TOTP) Provisioning, Federated Authentication Design, OAuth 2.0 Engine Customization, Role-Based Access Control (RBAC).
- **Application Security & DevSecOps:** Field-Level Encryption Engines, Cryptographic Persistence Layer Configuration, Secure Session Validation Hooks, Immutable Audit Log Trailing, API Rate Limiting, HTTP Protection Header Enforcement, Secure WebSocket Tunnels.
- **Software & Ecosystem Development:** Robust Backend Engineering using Python (3.10+), Flask Application Ecosystems, SQLAlchemy Object-Relational Mapping (ORM), Embedded SQL Databases (SQLite), WSGI Servers (Gunicorn), Reverse Proxy Deployment (Nginx), AI Model Integration, Frontend Layout Logic (HTML5, CSS3, JavaScript).
- **Defensive Testing & Vulnerability Validation:** Penetration-style API Probing, SQL Injection (SQLi) Structural Defense, Session Hijacking Countermeasures, Validation Drift Mitigation, Localized Credential Entropy Assessment.

PROFESSIONAL EXPERIENCE & RESEARCH INTERNSHIPS

IBM Remote / Institutional Partnership
Cybersecurity & Software Development Engineering Intern Jul 2024 – Aug 2025

- Analyzed industrial-grade enterprise technology patterns, building and debugging system infrastructure frameworks alongside experienced engineering mentors.
- Investigated modern runtime vulnerability metrics across distributed applications, evaluating security validation logs to mitigate architectural compliance failures.
- Researched local configuration flaws and active web stack vulnerabilities, drafting technical defense profiles to secure structural application endpoints.

Yenepoya Institute of Arts, Science, Commerce and Management Mangaluru, India
Cyber Security Research Intern – Identity Security Architecture Mar 2026 – Apr 2026

- Designed and prototyped an enterprise-grade identity platform focused on preserving data security properties for high-risk clinical facilities handling Protected Health Information (PHI).
- Constructed an isolated authentication engine separating user tiers (Administrative, Medical Staff, Patient) using customized token parsing middleware and rigid decorator validation patterns.
- Embedded cryptographic wrapper classes into the data layer, applying deterministic AES-256 field-level protection mechanisms to safeguard sensitive attributes directly inside persistent storage.

Yenepoya Institute of Arts, Science, Commerce and Management

Mangaluru, India

Network Security Research Intern – DDoS Mitigation Project

Sep 2024 – Nov 2024

- Documented systematic network traffic anomalies by assessing pcap stream records, evaluating packet behavior, and monitoring payload characteristics.
- Created threshold baseline models aimed at programmatically dropping malformed incoming protocol streams before backend computational channels faced capacity exhaustion.

Deloitte

Forge Virtual Training Portal

Cyber Security, Technology & Technical Analytics Simulator Graduate

2025 – 2026

- **Cyber Security Vector:** Conducted live tracking simulation evaluations covering active network scanning telemetry logs, incident profiling, and threat-containment reporting procedures.
- **Tech & Analytics Vector:** Structured multi-tier backend app relational mappings and deployed simulated auditing structures to capture live system telemetry flows cleanly.

ACHIEVEMENTS & COMPETITIONS

- **2nd Place – Capture The Flag (CTF) Arena — IBM ICE Day 2025:** Solved analytical exploitation challenges covering cryptanalysis, application code reversing, web system validation evasion, and network capture analysis.
- **2nd Runner-Up – Accelerated Hackathon Event — IBM ICE Day 2025:** Co-engineered a secure software tool under strict timeframe challenges, defending structural implementation choices in front of technical review panels.
- **Strategic Thought Leadership Industry Post:** Researched, drafted, and published a structural analysis evaluating technical infrastructure shifts, identity landscape modernizations, and protection priorities within India's growing digital ecosystem.

TECHNICAL PROJECTS PORTFOLIO

Ayurvedic Healthcare Identity Framework (Project PRJN26-161)

Lead Architect

Core Stack: Python, Flask, SQLite, Cryptography (Fernet), PyOTP, Authlib, Flask-SocketIO, Nginx, TLS 1.3

- Programmed an integrated zero-trust platform requiring conditional step-up token validations when users attempted high-risk database transactions or session lifetimes surpassed specific boundaries.
- Formulated immutable-style system audit logging trails tracking validation attempts, privileged elevation commands, and decryption requests to maintain absolute visibility.
- Evaluated data delivery behaviors over event-driven WebSockets, ensuring live telemetry updates occurred reliably without compromising cryptographic layer execution performance.

Future City – Smart Urban Planning Engine

Core Hackathon Developer

Platform: National Code Sprint Hackathon Competition

- Contributed to the layout development of an AI-driven smart city optimization engine, designing technical layouts capable of interpreting urban growth maps safely.

Glorius Gems – Web Development Inventory Platform

Lead Developer

Code Repository: github.com/Anantkamath00/GloriusGems

- Built an asset presentation catalog application using web backend logic, optimizing operational request speeds, session state management, and asset rendering structures.

Code Repository: github.com/Anantkamath00/AegisPass

- Developed a local computational utility designed to evaluate raw string compositions, analyzing password structural entropy blocks and calculating structural resistance against baseline wordlist attacks.

PROFESSIONAL CERTIFICATIONS

1. **Google Professional Cybersecurity Certificate:** Earned rigorous validation across foundational security fields, specializing in automated Python verification script handling, Linux terminal analysis, and network packet capture utilities.
2. **Managing Security and Administration in SAP Analytics Cloud:** Domain verification credential focusing on centralized identity handling models, fine-grained access policies, and enterprise management trace trails.